

Solutions

- 68% of nitric acid in water
 - Has zero enthalpy of mixing
 - Cannot be separated by distillation
 - Follows Raoult's Law
 - Has zero volume change of mixing
- Assuming complete dissociation, the value of van't Hoff factor for Na_2SO_4 is
 - 1
 - 2
 - 3
 - 4
- Consider a blood cell placed in a NaCl solution of concentration more than 0.9% (w/V). Which one of the following statements is true in this case?
 - Osmosis takes place from the cell to the NaCl solution, and the cell shrinks.
 - Osmosis takes place from the NaCl solution to the cell, and the cell swells.
 - Osmosis takes place from the cell to the NaCl solution, and the cell swells.
 - Osmosis takes place from the NaCl solution to the cell, and the cell shrinks.
- De-salination of sea water can be done by
 - Dialysis
 - Hypertonic solution
 - Evaporation
 - Reverse osmosis
- During osmosis, the solvent molecules are moving from
 - Hypertonic solution to isotonic solution
 - Hypertonic solution to hypotonic solution
 - Higher concentrated solution to lower concentrated solution
 - Hypotonic solution to hypertonic solution
- For a 0.1M solution, the colligative property will follow the order
 - $\text{NaCl} > \text{Na}_2\text{SO}_4 > \text{Na}_3\text{PO}_4$
 - $\text{NaCl} < \text{Na}_2\text{SO}_4 < \text{Na}_3\text{PO}_4$
 - $\text{NaCl} < \text{Na}_2\text{SO}_4 \approx \text{Na}_3\text{PO}_4$
 - $\text{NaCl} < \text{Na}_2\text{SO}_4 = \text{Na}_3\text{PO}_4$
- If abnormal molar mass is more than normal molar mass, the Van't Hoff factor (i) is
 - $i > 1$
 - $i = 1$
 - $i < 1$
 - $i \geq 1$
- In a solution containing a non-volatile solute, the mole fraction of solvent is 0.9. The relative lowering of vapor pressure is
 - 1
 - 0.1
 - 0.9
 - 1.1
- Isotonic solutions have the same
 - Surface tension
 - Vapour pressure
 - Osmotic pressure
 - Viscosity
- Pick out the ideal solution among the following
 - Phenol and Aniline
 - Ethanol and Acetone
 - Chloroform and Acetone
 - n-Hexane and n-Heptane
- The concentration term depends on temperature is
 - ppm
 - mole fraction
 - molality
 - molarity

12. The condition caused when scuba divers go into deep sea water is
(A) Anoxia (B) Hemophilia
(C) Bends (D) None
13. The K_H values for Ar(g) , $\text{CO}_2\text{(aq)}$, HCHO(aq) , and $\text{CH}_4\text{(aq)}$ are 40.39, 1.67, 1.83×10^{-3} , and 0.413 respectively. Arrange them in the order of their increasing solubility
(A) $\text{HCHO} < \text{CH}_4 < \text{CO}_2 < \text{Ar}$ (B) $\text{HCHO} < \text{CO}_2 < \text{CH}_4 < \text{Ar}$
(C) $\text{Ar} < \text{CO}_2 < \text{CH}_4 < \text{HCHO}$ (D) $\text{Ar} < \text{CH}_4 < \text{CO}_2 < \text{HCHO}$
14. The dissolution of a gas in a liquid is governed by
(A) Roul't's law (B) Henry's law
(C) Dalton's law of Pressure (D) Boyle's law
15. The maximum amount of a solid solute that can be dissolved in a specific amount of a given liquid solvent does not depend upon
(A) Nature of solvent (B) Pressure
(C) Nature of solute (D) Temperature
16. The mixture which shows positive deviation from Raoult's law is
(A) Ethanol + Acetone (B) Benzene + Toluene
(C) Acetone + Chloroform (D) Chlorobenzene + Bromoethane
17. The most diluted solution is (molar mass of $\text{NaCl} = 58.5 \text{ g mol}^{-1}$)
(A) 2% (w/w) NaCl in water (B) 1M NaCl
(C) 2 ppm of NaCl (D) 2 m NaCl
18. The pressure cooker reduces the cooking timing because
(A) Heat is more evenly distributed
(B) The boiling point of water inside the cooker is elevated
(C) The boiling point of water inside the cooker is depressed
(D) Higher pressure inside the cooker tenderizes food
19. The solubility of a gas in water depends on
(A) Nature of the gas (B) Temperature
(C) Pressure of the gas (D) All of the above
20. The solubility of a solid gas does not depend on
(A) Nature of the solute (B) Nature of the solvent
(C) Pressure (D) None of these
21. The solution which has lower osmotic pressure with respect to another is called
(A) Isotonic solution (B) Hypertonic solution
(C) Hypotonic solution (D) None of these
22. The solutions which show a large positive deviation from Raoult's law form
(A) Minimum boiling point azeotrope
(B) Maximum boiling point azeotrope
(C) Minimum or maximum boiling point azeotrope
(D) No azeotrope
23. The tanks used by most scuba divers are filled with air diluted with helium of around
(A) 88.3% (B) 56.2% (C) 32.1% (D) 11.7%

24. The value of Henry's constant K_H is
(A) Greater for gases with higher solubility.
(B) Greater for gases with lower solubility.
(C) Constant for all gases.
(D) Not related to the solubility of gases.
25. Value of Henry's constant (K_H)
(A) Increases with increase in temperature
(B) Decreases with increase in temperature
(C) Remains constant
(D) First increases, then decreases
26. When a substance is dissolved in a solvent, the vapor pressure for the solvent is decreased. This results in
(A) The solution having a lower osmotic pressure than the solvent
(B) A decrease in the boiling point of the solvent
(C) The solution having a higher freezing point than the solvent
(D) An increase in the boiling point of the solution
27. Which of the following aqueous solutions should have the highest boiling point?
(A) 1.0 M NaOH (B) 1.0 M Na_2SO_4
(C) 1.0 M NH_4NO_3 (D) 1.0 M KNO_3
28. Which of the following liquid-liquid solutions show negative deviation from Raoult's law?
(A) Benzene and Toluene (B) Water and Ethanol
(C) Acetone and Chloroform (D) Chlorobenzene and Bromobenzene
29. Which of the following solutions is used as an antifreezing agent in car radiators to cool the engine?
(A) 35% (v/v) solution of ethylene glycol
(B) 35% (m/m) solution of ethylene glycol
(C) 35% (m/v) solution of ethylene glycol
(D) 35% (v/m) solution of ethylene glycol
30. Which one of the following is not a solid solution?
(A) Hydrogen in palladium (B) Camphor in nitrogen gas
(C) Amalgam of mercury & sodium (D) Copper in gold

Electrochemistry:

31. A galvanic cell has an electrical potential of 1.1 V. If an opposing potential of 1.1 V is applied to this cell, what will happen to the cell reaction and current flowing through the cell?
- (A) The reaction stops and no current flows through the cell.
(B) The reaction continues but current flows in the opposite direction.
(C) The concentration of reactants becomes unity and current flows from cathode to anode.
(D) The cell does not function as a galvanic cell and zinc is deposited on the zinc plate.
32. An electrochemical cell can behave like an electrolytic cell when
- (A) $E_{\text{cell}} = 0$ (B) $E_{\text{cell}} > E_{\text{ext}}$ (C) $E_{\text{cell}} < E_{\text{ext}}$ (D) $E_{\text{cell}} = E_{\text{ext}}$
33. Assertion: The specific conductivity decreases with dilution.
Reason: The number of ions per unit volume that carry the current in a solution decreases on dilution.
- (A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.
(B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.
(C) Assertion is true, but the reason is false.
(D) Both assertion and reason are false.
34. E_{cell} of Daniel cell is calculated by using the following formula
- (A) $E_{\text{cell}} = E^{\circ}_{\text{cell}} - 0.0295 \log (\text{Cu}^{2+}/\text{Zn}^{2+})$
(B) $E_{\text{cell}} = E^{\circ}_{\text{cell}} - 0.0295 \log (\text{Cu}/\text{Zn}^{2+})$
(C) $E_{\text{cell}} = E^{\circ}_{\text{cell}} - 0.0295 \log (\text{Zn}^{2+}/\text{Cu}^{2+})$
(D) $E_{\text{cell}} = E^{\circ}_{\text{cell}} - 0.0295 \log (\text{Cu}^{2+}/\text{Zn})$
35. Fluorine has the highest standard electrode potential, indicating that fluorine gas is
- (A) Strongest reducing agent (B) Strongest oxidizing agent
(C) Weak oxidizing agent (D) SEP value is zero
36. If the E°_{cell} for a given reaction has a negative value, then which of the following is the correct relationship for the value of ΔG° and K_{eq} ?
- (A) $\Delta G^{\circ} > 0$ and $K_{\text{eq}} > 1$ (B) $\Delta G^{\circ} < 0$ and $K_{\text{eq}} < 1$
(C) $\Delta G^{\circ} > 0$ and $K_{\text{eq}} < 1$ (D) $\Delta G^{\circ} < 0$ and $K_{\text{eq}} > 1$
37. In a galvanic cell
- (A) Chemical energy is converted into heat
(B) Electrical energy is converted into heat
(C) Chemical energy is converted into electrical energy
(D) Electrical energy is converted into chemical energy

38. Match the batteries in Column I with their anode material in Column II.

Column I

Column II

- (i) Leclanché cell (a) Lead (Pb)
(ii) Mercury cell (b) Hydrogen(H_2) gas
(iii) Lead storage battery (c) Zinc (Zn)
(iv) H_2 - O_2 fuel cell. (d) Zinc – Mercury amalgam($Zn(Hg)$)
(A) i-a, ii-b, iii-c, iv-d (B) i-b, ii-a, iii-d, iv-c
(C) i-c, ii-d, iii-a, iv-b (D) i-d, ii-c, iii-a, iv-b
39. Saturated solution of KNO_3 is used to make a salt bridge because
(A) Velocity of K^+ is greater than that of NO_3^-
(B) Velocity of NO_3^- is greater than that of K^+
(C) Velocity of both K^+ and NO_3^- are nearly the same
(D) KNO_3 is highly soluble in water
40. Statement I: Conductivity (κ) of a solution decreases with dilution.
Statement II: Molar conductivity (λ_m) of solution increases with dilution.
Identify the correct statement
A. Both statements I and II are correct
B. Both statements I and II are incorrect
C. Statement I is correct but statement II is incorrect
D. Statement I is incorrect but statement II is correct
41. Statement I: Cu displaces H_2 gas from dilute acids.
Statement II: ions Cu^{2+} get reduced more easily than H^+ ions
(A) Both Statement I and II are correct
(B) Both Statement I and II are incorrect
(C) Statement I is correct and Statement II is incorrect
(D) Statement I is incorrect and Statement II is correct
42. The E° values of Fe^{2+} / Fe , Zn^{2+} / Zn , and Sn^{2+} / Sn are -0.44V, -0.76V, and -0.14V respectively. Which of these metals can displace iron from an Fe^{2+} solution?
(A) Both Zn and Sn (B) Zn only
(C) Sn only (D) Neither Zn nor Sn
43. The molar conductivity is maximum for the solution of concentration
(A) 0.005 M (B) 0.001 M (C) 0.004 M (D) 0.002 M
44. The product formed at the anode during the electrolysis of aqueous NaCl is
(A) H_2 (B) O_2 (C) Na (D) Cl_2

Application Based Questions:

✉ learn@simplifiedminds.com

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45. The quantity of electricity required for the reduction of one mole of Al^{3+} ions is
(A) 1 Faraday (B) 2 Faraday (C) 3 Faraday (D) 4 Faraday
46. What will happen in the Daniel cell, when $E_{\text{ext}} < 1.1 \text{ V}$?
(A) Electrons flow from Zn rod to Cu rod; hence current flows from Cu to Zn
(B) Electrons flow from Cu rod to Zn rod; hence current flows from Cu to Zn
(C) Electrons flow from Zn rod to Cu rod; hence current flows from Zn to Cu
(D) Electrons flow from Cu rod to Zn rod; hence current flows from Zn to Cu
47. When one Faraday current is passed, which of the following would deposit one-gram atomic mass of the metal?
(A) NaCl (B) ZnSO_4 (C) AlCl_3 (D) BaCl_2
48. Which of these statements about a galvanic cell are not true?
i) The cathode carries a positive sign.
ii) Oxidation occurs at the anode.
iii) Electrons flow from cathode to anode in the external circuit.
iv) The anode carries a negative sign
(A) i and iii (B) i and ii (C) ii and iii (D) ii and iv

Chemical Kinetics:

49. Acid hydrolysis of ethyl acetate is an example of
- (A) Zero order reaction (B) First order reaction
(C) Second order reaction (D) Pseudo first order reaction
50. All natural and artificial radioactive decay of unstable nuclei take place by
- (A) Zero order kinetics (B) Half-order kinetics
(C) First-order kinetics (D) Second-order kinetics
51. For $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$, the following initial-rate data were obtained. Choose the rate equation.
- (A) $\text{Rate} = k[\text{NO}]^2$ (B) $\text{Rate} = k[\text{NO}]^2 [\text{O}_2]^2$
(C) $\text{Rate} = k[\text{NO}] [\text{O}_2]$ (D) $\text{Rate} = k[\text{NO}]^2 [\text{O}_2]$
52. Consider the Arrhenius equation $k = e^{-E_a/RT}$ and mark the correct option
- (A) Rate constant increases exponentially with increasing activation energy and decreasing temperature.
(B) Rate constant decreases exponentially with increasing activation energy and decreasing temperature.
(C) Rate constant increases exponentially with increasing activation energy and increasing temperature.
(D) Rate constant increases exponentially with decreasing activation energy and decreasing temperature.
53. For a chemical reaction $\text{A} \rightarrow \text{B}$, it is found that the rate of a reaction doubles when the concentration of A is doubled. The order of the reaction is
- (A) One (B) Two (C) Zero (D) Three
54. For a zero-order reaction, the intercept for the plot of concentration of reactants vs. time is equal to
- (A) $-k$ (B) $\log k$ (C) $[\text{R}_0]$ (D) $\log [\text{R}_0]$
55. In a reaction $\text{A} + \text{B} \rightarrow \text{products}$, rate is doubled when the concentration of B is doubled, and rate increases by a factor of 8 when the concentrations of both the reactants (A and B) are doubled. The rate law for the reaction can be written as:
- (A) $\text{Rate} = K [\text{A}][\text{B}]^2$
(B) $\text{Rate} = K [\text{A}]^2[\text{B}]^2$
(C) $\text{Rate} = K [\text{A}]^2[\text{B}]$

(D) $\text{Rate} = K [\text{A}] [\text{B}]$

56. In a reaction $\text{Hg} + \text{Cl}_2 \rightarrow \text{HgCl}_2$, the rate of disappearance of Hg is equal to
(A) Rate of disappearance of Cl_2 (B) Rate of appearance of HgCl_2
(C) Rate of appearance of Cl_2 (D) Both (a) and (b)
57. In a reaction, when the concentration of a reactant is increased two times, the increase in the rate of reaction was four times. The order of the reaction is
(A) First order (B) Zero order
(C) Pseudo first order (D) Second order
58. In presence of a catalyst, the heat evolved or absorbed during the reaction
(A) Increases (B) Decreases
(C) Depends on the reaction (D) Remains unchanged
59. Order of a reaction in which unit of rate of reaction and rate constant are the same
(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) 2
60. Statement I: A catalyst does not alter Gibb's energy (ΔG) of a reaction.
Statement II: A catalyst alters the activation energy (E_a) of a reaction by providing an alternate pathway.
(A) Both Statement I and II are correct
(B) Both Statement I and II are incorrect
(C) Statement I is correct, but Statement II is incorrect
(D) Statement I is incorrect, but Statement II is correct
61. Statement I: If in a zero-order reaction, the concentration of the reactant is doubled, the half-life period is also doubled.
Statement II: For a first-order reaction, the half-life period is independent of the initial concentration of the reactant.
(A) Both Statement I and II are correct
(B) Both Statement I and II are incorrect
(C) Statement I is correct but Statement II is incorrect
(D) Statement I is incorrect but Statement II is correct
62. The collision frequency in a reaction depends on
(A) Number of moles of reactants
(B) Proper orientation
(C) Activated reactant molecules
(D) Both (b) and (c)

63. The following factor which does not affect the rate of reaction is
 (A) Molecularity (B) Temperature
 (C) Catalyst (D) Concentration of reactant
64. The half-life period of a first-order reaction is 15 minutes. The amount of substance left after one hour will be
 (A) One-half (B) One-fourth (C) One-eighth (D) One-sixteenth
65. The molecularity of a reaction cannot be
 (A) Zero (B) One
 (C) Both (a) and (b) (D) Whole number
66. The number of collisions per second per unit volume of the reaction is known as
 (A) Collision energy (B) Activation energy
 (C) Collision frequency (D) Kinetic energy
67. The order of a chemical reaction having the rate expression, $\text{Rate} = k[A][B]$ is
 (A) 1 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
68. The overall order of a reaction which has the rate expression $\text{Rate} = K[A]^{1/2}[B]^{3/2}$ is
 (A) $\frac{1}{2}$ (B) 2 (C) $\frac{3}{2}$ (D) 1
69. The order for a reaction with the rate law $\text{Rate} = k[A]^1[B]^1$ is:
 (A) Zero (B) Fraction (C) One (D) Two
70. The rate constant of a reaction is $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$. The order of the reaction is
 (A) Zero (B) One (C) Two (D) Three
71. The S.I. unit of the rate constant of a first-order reaction is
 (A) s^{-1} (B) mol dm^{-3} (C) $\text{mol dm}^{-3} \text{ s}^{-1}$ (D) $\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$
72. The slope of the Arrhenius plot ($\log k$ versus $1/T$) for a given reaction is found to be -2000 K . What is the activation energy (E_a) of the reaction in J mol^{-1} ? (Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)
 (A) $38,314.5 \text{ J mol}^{-1}$ (or 38.3 kJ mol^{-1}) (B) $19,157.2 \text{ J mol}^{-1}$
 (C) $76,629.0 \text{ J mol}^{-1}$ (D) $8,314.0 \text{ J mol}^{-1}$
73. The time required for 100 percent completion of a zero-order reaction is
 (A) $2k / [R]_0$ (B) $[R]_0 / 2k$ (C) $[R]_0 / k$ (D) $[R]_0 k$
74. The unit of the rate constant of a reaction is the same as that of its rate. The order of the reaction is
 (A) Zero order. (B) First order. (C) Second order. (D) Pseudo order.
75. What is the value of 'x' if the total order of the reaction, which has the rate expression:
 $\text{Rate} = k[A]^x[B]^{2x}$ is $3/2$?
 (A) $\frac{1}{2}$ (B) $3/2$ (C) 4 (D) 1

Application Based Questions:

76. When the initial concentration of the reactant is doubled, then the half-life of a zero-order reaction

- (A) Is halved (B) Is doubled (C) Is tripled (D) Remains unchanged

77. Which of the following factors does not affect the rate of reaction?

- (A) Molecularity (B) Temperature
(C) Catalyst (D) Concentration of reactant

78. Which of the following is correct for a zero-order reaction?

- (A) $t_{1/2} \propto [R_0]^0$ (B) $t_{1/2} \propto [R_0]^2$ (C) $t_{1/2} \propto [R_0]^{-1}$ (D) $t_{1/2} \propto [R_0]^1$



SIMPLIFIED MINDS

D and F Block elements:

79. Actinoids are
(A) Radioactive (B) Prepared in nanograms
(C) Having smaller half-lives (D) All the above are correct
80. Among ScCl_3 , TiCl_3 , and ZnCl_2 , the colored compound is
(A) ScCl_3 (B) TiCl_3 (C) ZnCl_2 (D) All the above
81. Diamagnetic compound among the following is
(A) Cu_2O (B) MnO_2 (C) V_2O_3 (D) FeO
82. Identify the correct sequence of the number of unpaired electrons of the following ions.
(A) $\text{Ti}^{3+} > \text{Cr}^{3+} > \text{Fe}^{3+} > \text{Ni}^{2+}$
(B) $\text{Fe}^{3+} > \text{Ni}^{2+} > \text{Ti}^{2+} > \text{Cr}^{3+}$
(C) $\text{Fe}^{3+} > \text{Cr}^{3+} > \text{Ni}^{2+} > \text{Ti}^{3+}$
(D) $\text{Fe}^{3+} > \text{Cr}^{3+} > \text{Ti}^{3+} > \text{Ni}^{2+}$
83. In a neutral or faintly alkaline medium, KMnO_4 oxidizes iodide ions into iodate. The change in oxidation state of manganese in this reaction is from
(A) +7 to +2 (B) +7 to +4 (C) +6 to +4 (D) +7 to +6
84. In aqueous solution, copper (I) ions undergo
(A) Oxidation (B) Reduction
(C) Disproportionation (D) Redox reaction
85. Potassium permanganate oxidizes sulphurous acid (H_2SO_3) into
(A) S (B) $\text{S}_2\text{O}_7^{2-}$ (C) SO_4^{2-} (D) SO_3^{2-}
86. The aqueous solution of a transition metal ion M^+ is colorless. The ion M^+ is
(A) Mn^{2+} (B) Cr^{3+} (C) V^{4+} (D) Ti^{4+}
87. The characteristic feature of transition metal is
(A) They show variable oxidation state
(B) They do not form complexes
(C) They are not good catalysts
(D) They exhibit only diamagnetism
88. The correct order of magnetic moment is
(A) $\text{Sc}^{3+} < \text{Ti}^{3+} < \text{Ti}^{2+}$ (B) $\text{Ti}^{3+} < \text{Ti}^{2+} < \text{Sc}^{3+}$
(C) $\text{Ti}^{2+} < \text{Sc}^{3+} < \text{Ti}^{3+}$ (D) $\text{Sc}^{3+} < \text{Ti}^{2+} < \text{Ti}^{3+}$
89. The correct order of melting point is

- (A) $\text{Cr} > \text{Mn} > \text{Fe}$ (B) $\text{Fe} > \text{Mn} > \text{Cr}$
(C) $\text{Cr} > \text{Fe} > \text{Mn}$ (D) $\text{Mn} > \text{Fe} > \text{Cr}$
90. The electronic configuration of Cu (II) is $3d^9$, whereas Cu(I) is $3d^{10}$. Which of the following is correct?
(A) Cu (II) is less stable.
(B) Cu (II) is more stable.
(C) Stability of both depends upon the nature of their salts.
(D) Cu(I) and Cu (II) are equally stable.
91. The magnetic moment of an element is 1.73 BM. The number of unpaired electrons in it is
(A) 2 (B) 3 (C) 1 (D) 0
92. The most stable manganese compound is
(A) Mn_2O_7 (B) MnF_4 (C) MnO_2 (D) Mn_2SO_4
93. The number of moles of $\text{K}_2\text{Cr}_2\text{O}_7$ reduced by one mole of Sn^{2+} ion is
(A) $1/3$ (B) 3 (C) $1/6$ (D) 6
94. The set of 3d series elements that show anomalous electronic configurations are
(A) Cr and Mn (B) Cu and Zn
(C) Cr and Cu (D) Mn and Zn
95. The transition metal present in bronze is
(A) Zinc (B) Tin (C) Silver (D) Copper
96. The transition metal present in the red pigment of blood haemoglobin is
(A) cobalt (B) Nickel (C) Iron (D) copper
97. Which of the following actinoids show oxidation states up to +7?
(A) Am (Americium) (B) Th (Thorium)
(C) U (Uranium) (D) Np (Neptunium)
98. Which of the following ions has the least magnetic moment?
(A) Cu^{2+} (B) Co^{3+} (C) Ni^{2+} (D) Fe^{2+}
99. Which of the following is a colorless ion?
(A) V^{3+} (B) Fe^{3+} (C) Ti^{3+} (D) Zn^{2+}
100. Which of the following pairs have the same paramagnetic moment?
(Atomic numbers: Ti = 22, Mn = 25, Ni = 28, Cu = 29)
(A) Ti^{3+} , Ni^{2+} (B) Cu^{2+} , Ti^{3+} (C) Ti^{2+} , Cu^{2+} (D) Mn^{2+} , Cu^{2+}

101. Which of the following statements about transition elements is incorrect?

- (A) They show variable oxidation states.
- (B) All the ions are colored.
- (C) They exhibit diamagnetic and paramagnetic properties.
- (D) They exhibit catalytic properties.

102. Which of the following transition metal ions is colorless?

- (A) V^{2+} (B) Cr^{3+} (C) Zn^{2+} (D) Ti^{3+}

Haloalkanes and Haloarenes:

103. A mixture of chloromethane and chlorobenzene are treated with sodium in dry ether, the product obtained is

- A. Ethane B. Toluene C. Diphenyl D. All the above

104. Among the following alkyl halides, which is more reactive towards SN_2 reaction?

- A. R-I B. R-Cl C. R-F D. R-Br

105. Assertion: SN_2 reactions do not proceed with retention.

Reason: SN_2 reaction proceeds in single steps.

Identify the correct statement

- A. If both assertion and reason are correct and reason is correct explanation for assertion
- B. If both assertion and reason are correct but reason is not correct explanation for assertion
- C. If assertion is correct but reason is incorrect
- D. If assertion is incorrect but reason is correct

106. Chlorobenzene reacts with magnesium in dry ether to give a compound A, which is

- (A) C_6H_5OH (B) C_6H_5MgCl (C) $C_2H_5CH_2MgCl$ (D) $MgCl_2$

107. Chloroform is stored completely filled in a closed dark-colored bottle. This is to

- (A) Prevent the oxidation of chloroform by air in the presence of sunlight.
- (B) Prevent the evaporation of chloroform.
- (C) Prevent the reaction of chloroform with nitrogen in the air.
- (D) Prevent the reaction of chloroform with CO_2 to form extremely poisonous gas carbonyl chloride.

108. Four haloalkane compounds represented by the letters M, N, O, and P having boiling points of $24.2^\circ C$, $38^\circ C$, $3.56^\circ C$, and $101.6^\circ C$ respectively. Among the four compounds, N is most likely to be

- (A) CH_3Cl (B) CH_3Br (C) C_2H_5Br (D) C_3H_7I

109. Freon can be prepared by

- (A) Finkelstein reaction
- (B) Wurtz reaction
- (C) Friedel-Crafts reaction
- (D) Swartz reaction

110. In vinylic halides, the halogen atom is attached to a carbon atom with
 (A) sp^3 hybridization (B) sp^2 hybridization
 (C) sp hybridization (D) dsp^2 hybridization
111. Match the organic compounds in Column I with their correct applications in Column II:
- | Column I (Compound) | Column II (Application) |
|--|-------------------------|
| (1) Carbon tetrachloride (CCl_4) | (A) Fire extinguisher |
| (2) Iodoform (CHI_3) | (B) Antiseptic |
| (3) Chloroform ($CHCl_3$) | (C) Solvent for fats |
| (4) Methylene chloride (CH_2Cl_2) | (D) Paint remover |
| (A) (1)-(A), (2)-(B), (3)-(C), (4)-(D) | |
| (B) (1)-(D), (2)-(C), (3)-(B), (4)-(A) | |
| (C) (1)-(B), (2)-(A), (3)-(D), (4)-(C) | |
| (D) (1)-(C), (2)-(D), (3)-(A), (4)-(B) | |
112. $R - X + NaI$ (dry acetone) $\rightarrow RI + NaX$, name the reaction
 (A) Wurtz reaction (B) Fittig reaction
 (C) Swarts reaction (D) Finkelstein reaction
113. Select the incorrect match
 A. Swartz reaction - CoF_2 B. Finkelstein reaction - NaI
 C. Sandmeyer's reaction - $CuCl_2$ D. Wurtz reaction - Na / dry ether
114. Tertiary alkyl halides are least reactive towards SN_2 reaction because of
 A. Inductive effect B. Steric effect
 C. Resonance effect D. Electrometric effect
115. The best reagent used to produce pure alkyl halides from alcohol is
 A. Concentrated HCl B. Concentrated HBr
 C. $SOCl_2$ D. $NaBr$
116. The chemical formula of phosgene is
 A. $SOCl_2$ B. CH_2Cl_2 C. $COCl_2$ D. CCl_4
117. The chemical name of phosgene is
 A. Acetyl chloride B. Methyl chloride
 C. Carbonyl chloride D. Chloroform
118. The chiral molecule among the following is
 A. 1-Bromo butane B. 2-Bromobutane

- C. 2-Bromo propane D. 2-Bromopropan-2-ol
119. The compound $\text{H}_2\text{C}=\text{CH}-\text{Cl}$ is an example of
 A. Allylic halides B. Aryl halides
 C. Benzylic halides D. Vinylic halides
120. The increasing order of boiling points of alkyl halides is
 A. $\text{RF} > \text{RCl} > \text{RBr} > \text{RI}$ B. $\text{RF} > \text{RI} > \text{RCl} > \text{RBr}$
 C. $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$ D. $\text{RCl} > \text{RBr} > \text{RF} > \text{RI}$
121. The IUPAC name of ethylidene chloride is
 A. 1-Chloroethene B. 1-Chloroethyne
 C. 1,2-Dichloroethane D. 1,1-Dichloroethane
122. The reaction $\text{CH}_3\text{Br} + \text{AgF} \rightarrow \text{CH}_3\text{F} + \text{AgBr}$ is called
 A. Swarts reaction B. Wurtz reaction
 C. Finkelstein reaction D. Gattermann reaction
123. When 2-Bromo pentane is treated with alcoholic potash, the major product produced is
 A. 1-Pentene B. 2-Pentene C. 2-Pentane D. 3-Pentene
124. Which of the following alkyl halides will undergo SN_1 reaction most readily?
 A. $(\text{CH}_3)_3\text{C}-\text{F}$ B. $(\text{CH}_3)_3\text{C}-\text{Cl}$ C. $(\text{CH}_3)_3\text{C}-\text{Br}$ D. $(\text{CH}_3)_3\text{C}-\text{I}$

Alcohols, Phenols and Ethers:

125. A compound P with formula C_3H_6 on reaction with H_2O in presence of H_2SO_4 gives the compound Q. Compound Q on heating with Cu at 573K produces the compound R. Compounds P, Q, and R are
 A. P = Propan-1-ol, Q = Propene, R = Propanone
 B. P = Propene, Q = Propan-2-ol, R = Propene
 C. P = Propene, Q = Propan-2-ol, R = Propanone
 D. P = Propanone, Q = Propan-1-ol, R = Propane
126. An organic compound 'A' containing C, H, and O has a pleasant odor with a boiling point of 78°C . On boiling 'A' with conc. H_2SO_4 , a colorless gas is produced which decolorizes bromine water and alkaline KMnO_4 . The organic compound A is
 A. $\text{C}_2\text{H}_5\text{Cl}$ B. $\text{C}_2\text{H}_5\text{OH}$ C. $\text{C}_2\text{H}_5\text{COOCH}_3$ D. C_6H_6
127. By which test can 1° , 2° , and 3° alcohols be detected?
 A. Hinsberg test B. Tollen's test C. Lucas test D. All of these
128. Byproduct obtained during preparation of phenol from cumene is

- A. Ketone B. Aldehyde C. Alcohol D. Cresol
129. Enantiomers only differ with respect to
A. Melting point B. Boiling point
C. Refractive index D. The rotation of plane-polarized light
130. Ethane-1,2-diol is an example of
A. Monohydric alcohol B. Dihydric alcohol
C. Dihydric phenol D. Monohydric phenol
131. Find out the product of the following reaction:
(Reaction involving NaOH, 623K, 300 atm)
A. Aromatic amine
B. Aromatic hydroxyl compound
C. Aromatic dihydroxy compound
D. Aromatic nitro compound
132. Halogenation of benzene occurs in the presence of Lewis acid, but halogenation of phenol occurs even in the absence of Lewis acid due to
A. Highly deactivating effect of -OH group attached to the benzene ring
B. Highly activating effect of -OH group attached to the benzene ring
C. Highly deactivating effect of the benzene ring
D. Highly activating effect of the benzene ring
133. How many alcohols with molecular formula $C_4H_{10}O$ are chiral in nature?
A. 1 B. 2 C. 3 D. 4
134. Identify the simple ether from the following
A. $CH_3OC_2H_5$ B. $C_2H_5OC_6H_5$ C. $C_6H_5OC_6H_5$ D. $C_6H_5OCH_3$
135. In Kolbe's reaction, phenol is converted into
A. Salicylic acid B. Salicylaldehyde C. Benzoic acid D. Benzaldehyde
136. In picric acid
A. Three -OH groups are present
B. Two -OH groups are present
C. Three -NO₂ groups are present
D. One -NO₂ group is present
137. O-nitrophenol and p-nitrophenol are separated by
A. Sublimation B. Fractional distillation
C. Crystallization D. Steam distillation
138. Phenol is less acidic than
A. o-Nitrophenol B. Ethanol
C. o-Methylphenol D. o-Methoxyphenol
139. Phenol is more acidic than

- A. Ethanol
C. m-Nitrophenol
- B. o-Nitrophenol
D. p-Nitrophenol
140. Phenol on oxidation with chromic acid gives
- A. Benzoquinone
C. Hydroquinone
- B. Catechol
D. Phthalic acid
141. Salicylaldehyde can be prepared by
- A. Finkelstein reaction
C. Reimer-Tiemann reaction
- B. Dow's process
D. Kolbe's reaction
142. The boiling points of alcohols increase with
- A. Increase in the number of C-atoms
B. Decrease in van der Waals forces
C. Increase of branching in the C-chain
D. Decrease in the surface area of the C-chain
143. The C-O bond angles of P, Q, and R are found to be 111.7° , 109° , and 108.9° , respectively. Compound P, Q, and R are
- A. P = Phenol, Q = Methanol, R = Methoxy Methane
B. P = Methoxy Methane, Q = Methanol, R = Phenol
C. P = Methanol, Q = Phenol, R = Methoxy Methane
D. P = Methoxy Methane, Q = Phenol, R = Methanol
144. The chemicals used to convert commercial alcohol into denatured alcohol are
- A. CuSO_4 and Pyrolidine
C. CuSO_4 and Pyridine
- B. Cu_2S and Pyrolidine
D. Cu_2S and Pyridine
145. The compound that reacts fastest with Luca's reagent at room temperature is
- A. 2-Methylpropan-2-ol
B. Butan-2-ol
C. 2-Methylpropan-1-ol
D. Butan-1-ol
146. The correct order of acid strength of the following compounds is:
- (i) o-bromophenol
(ii) p-bromophenol
(iii) picric acid (2,4,6-trinitrophenol)
(iv) 2,4,6-tribromophenol
- A. (iii) > (iv) > (i) > (ii)
C. (iv) > (iii) > (i) > (ii)
- B. (iii) > (iv) > (ii) > (i)
D. (i) > (ii) > (iv) > (iii)

147. The decreasing order of boiling points of ethanol, propane, and dimethyl ether is
A. Ethanol > Dimethyl ether > Propane
B. Ethanol > Propane > Dimethyl ether
C. Propane > Dimethyl ether > Ethanol
D. Propane > Ethanol > Dimethyl ether
148. The high boiling point of ethanol [$\text{C}_2\text{H}_5\text{OH}$] compared to methoxy methane [CH_3OCH_3] and propane is due to
A. Intra-molecular hydrogen bonding
B. Inter-molecular hydrogen bonding
C. Both (A) and (B)
D. None of these
149. The major byproduct during the preparation of phenol from cumene is
A. Acetone. B. Cresol. C. Alcohol. D. Acetaldehyde.
150. The major product formed when phenol is treated with excess bromine water is
A. o-Bromophenol B. p-Bromophenol
C. Picric acid D. 2,4,6-Tribromophenol
151. The role of CO_2 in Kolbe's reaction is
A. acts as catalyst B. act as nucleophile
C. act as weak electrophile D. act as strong electrophile.
152. What is the correct order of boiling points of alcohols having the same number of carbon atoms?
A. $1^\circ > 2^\circ > 3^\circ$ B. $3^\circ > 2^\circ > 1^\circ$
C. $3^\circ > 1^\circ > 2^\circ$ D. $2^\circ > 1^\circ > 3^\circ$
153. When an alcohol is treated with Luca's reagent, if immediate turbidity appears, then it confirms the presence of
A. Primary alcohol B. Secondary alcohol
C. Tertiary alcohol D. All of these
154. When phenol is heated with chloroform in sodium hydroxide followed by acidification, the product is
A. Chlorobenzene B. Benzyl chloride
C. Salicylaldehyde D. Salicylic acid
155. When phenol is heated with Zn dust, it gives
A. Benzyl alcohol B. Ethanol

- C. Methanol D. Benzene
156. When phenol is treated with bromine water, a white precipitate of the product obtained is
- A. 2-Bromophenol
B. 4-Bromophenol
C. Both 2-Bromophenol and 4-Bromophenol
D. 2,4,6-Tribromophenol

Coordination Compounds:

157. According to the spectrochemical series, the strongest field ligand is
- A. F^- B. NH_3 C. H_2O D. CO
158. According to Werner's theory, the primary valencies of the central atom are
- A. Satisfied by negative ions or neutral molecules
B. Satisfied by negative ions
C. Are equal to coordination number
D. Decide the geometry of the complex
159. Among the following statements, which statement is correct about metal carbonyls?
- A. Metal carbonyls are s-bonded organometallic compounds
B. Metal carbonyls are p-bonded organometallic compounds
C. Metal carbonyls involve both s and p bonds between metal and oxygen of the carbonyl group
D. Metal carbonyls involve both s and p bonds between metal and carbon of the carbonyl group
160. An ambidentate ligand is
- A. Cl^- B. CN^- C. OH^- D. NO_2^+
161. An example of an ambidentate ligand is
- A. CO B. CN^- C. NO_3^- D. SO_4^{2-}
162. Cobalt (III) chloride forms several octahedral complexes with ammonia. Which of the following will not give a test for chloride ion with silver nitrate solution?
- A. $CoCl_3 \cdot 4NH_3$ B. $CoCl_3 \cdot 5NH_3$
C. $CoCl_3 \cdot 6NH_3$ D. $CoCl_3 \cdot 3NH_3$
163. If $\Delta_o < P$ (Pairing energy), for the d_4 ion, an entry of the 4th electron gives the configuration
- A. $t_{2g}^2 e_g^2$ B. $t_{2g}^4 e_g^0$ C. $t_{2g}^3 e_g^1$ D. $t_{2g}^2 e_g^2$

164. In coordination compound $[\text{CoCl}_2(\text{en})_2]\text{Br}$, the primary and secondary valencies of Co are
A. 4 and 1 B. 6 and 1 C. 2 and 3 D. 3 and 6
165. In $\text{Fe}(\text{CO})_5$, Fe-C bond processes involve
A. π -character only. B. Both σ - and π -characters.
C. Ionic character. D. σ -character.
166. In metal carbonyls, the M-C sigma bond is formed by the donation of a lone pair of electrons on the
A. Carbonyl carbon atom to filled orbital of metal
B. Carbonyl carbon atom to vacant orbital of metal
C. Filled d-orbital of metal to vacant antibonding π orbital of CO
D. Filled d-orbital of metal to vacant bonding π orbital of CO
167. IUPAC name of $[\text{CoCl}(\text{en})_2\text{ONO}] \text{Br}$ is
A. ChloridoethylenediaminenitroCobalt (III) Bromide
B. ChloridoethylenediaminenitroCobaltate (III) Bromide
C. Chloridobis(ethylenediamine)nitritoCobaltate(III) Bromide
D. Chloridobis(ethylenediamine)nitritoCobalt (III) Bromide
168. Square planar complex of the type $[\text{MABXL}]$ (where A, B, X, L are unidentate) shows
A. Two Cis and one Trans
B. Two Cis and Two Trans
C. One Cis and two Trans
D. One Cis and One Trans
169. Square planar complex of the type MABXL (where A, B, X, L are unidentates) shows three isomers. Among them
A. Two cis and one trans
B. One cis and two trans
C. All three are cis only
D. All three are trans only
170. The bond between CO and metal in metal carbonyl is stronger due to
A. Synergic effect B. Chelating effect
C. Only σ bond effect D. Ambidentate nature of CO ligand
171. The central metal atoms/ions in coordination compounds are also called as
A. Lewis bases B. Brønsted acids
C. Brønsted bases D. Lewis acids
172. The complex $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ exhibits
A. Linkage isomerism B. Fac-mer isomerism

- C. Ionization isomerism D. Cis-trans isomerism
173. The complex $[\text{PtCl}_2.4\text{NH}_3\text{Br}_2]$ is treated with excess of AgNO_3 solution, and two moles of AgBr is precipitated. The primary and secondary valency of this complex is
A. 6 and 1 B. 6 and 2 C. 4 and 6 D. 3 and 6
174. The complex $\text{PtCl}_2.4\text{NH}_3\text{Br}_2$ is treated with excess of AgNO_3 solution, two mole of AgBr is precipitated. The primary and secondary valence of this complex is
A. 6 and 1 B. 6 and 2 C. 4 and 6 D. 3 and 6
175. The complexes $[\text{Co}(\text{NH}_3)_6]$ $[\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{NH}_3)_6]$ $[\text{Co}(\text{CN})_6]$ are examples of which type of isomerism?
A. Geometrical isomerism B. Ionization isomerism
C. Coordination isomerism D. Linkage isomerism
176. The electron configuration of an octahedral d^4 coordination entity with $\Delta_0 > P$ is
A. $t_2g^2 e_g$ B. $t_2g^3 e_g$ C. $t_2g^4 e_g$ D. $t_2g^2 e_g^2$
177. The geometry of the complex $\text{Fe}(\text{CO})_5$ is
A. octahedral B. tetrahedral
C. trigonal bipyramidal D. square pyramidal
178. The isomerism exhibited by complexes $[\text{Co}(\text{NH}_3)_5\text{SO}_4] \text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ is
A. Linkage isomerism B. Coordination isomerism
C. Ionisation isomerism D. Solvate isomerism
179. The IUPAC name of the complex ion $[\text{CoCl}_2(\text{en})_2]^+$ is
A. Di-chloridodiethylene diamine cobalt (III)
B. Dichlorido bis(ethane-1,2-diamine) cobalt (III)
C. Dichlorido bis(ethane-1,2-diamine) cobalt (IV)
D. Bis (ethane-1,2-diamine) dichlorido cobalt (III)
180. The IUPAC name of the ionization isomer of $[\text{Co}(\text{NH}_3)_5 \text{SO}_4] \text{Br}$ is
A. Penta-ammine sulphato cobalt (III) bromide
B. Penta-ammine bromido cobalt (III) sulphate
C. Penta-ammine sulphato cobalt (II) sulphate
D. Penta-ammine bromido sulphato cobaltate (III)
181. The magnitude of CFSE (Crystal Field Splitting Energy) (Δ_0) can be related to the configuration of d-orbitals in a coordination entity as
A. If $\Delta_0 < P$, the configuration is $t_2g^3 e_g$
B. If $\Delta_0 > P$, the configuration is $t_2g^6 e_g$
C. If $\Delta_0 > P$, the configuration is $t_2g^5 e_g$
D. If $\Delta_0 < P$, the configuration is $t_2g^4 e_g$
182. The magnitude of CFSE (Crystal Field Splitting Energy, Δ_0) can be related to the d^4 configuration of the central metal in an octahedral coordination entity as

- A. If $\Delta_0 < P$, the configuration is $t_{2g}^3 e_g$
 B. If $\Delta_0 > P$, the configuration is $t_{2g}^5 e_g$
 C. If $\Delta_0 > P$, the configuration is $t_{2g}^4 e_g$
 D. If $\Delta_0 < P$, the configuration is $t_{2g} e_g$
183. The oxidation number of Cu in $[\text{Cu}(\text{CN})_4]^{3-}$ is
 A. +1 B. +4 C. +2 D. +3
184. The oxidation state of metal in metal carbonyl complexes is
 A. 1 B. 2 C. 3 D. 0
185. The primary and secondary valencies of $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ complexes are
 $^{224}[\text{M}(\text{en})(\text{CO})] \text{Cl}$ is
 A. 3 and 3 B. 3 and 6 C. 6 and 3 D. 0 and 3
186. The primary and secondary valencies of iron in the complex $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ are
 A. 3 and 3 B. 6 and 6 C. 3 and 6 D. 6 and 3
187. The sum of coordination number and oxidation number of the metal M in the complex
 A. 6 B. 7 C. 8 D. 9
188. The terminal and bridged CO ligands in the compound $[\text{Co}_2(\text{CO})_8]$ are respectively
 A. 0, 2 B. 6, 1 C. 5, 2 D. 6, 2
189. Which of the following is an ambidentate ligand?
 A. CO B. NH_3 C. SCN^- D. Cl^-
190. Which of the following is the correct increasing order of strength in the spectrochemical series of ligands?
 A. $\text{EDTA}^{4-} < \text{C}_2\text{O}_4^{2-} < \text{Cl}^- < \text{OH}^- < \text{CN}^-$
 B. $\text{EDTA}^{4-} < \text{C}_2\text{O}_4^{2-} < \text{CN}^- < \text{OH}^- < \text{Cl}^-$
 C. $\text{Cl}^- < \text{C}_2\text{O}_4^{2-} < \text{Cl}^- < \text{OH}^- < \text{EDTA}^{4-}$
 D. $\text{Cl}^- < \text{OH}^- < \text{C}_2\text{O}_4^{2-} < \text{EDTA}^{4-} < \text{CN}^-$
191. Which of the following ligands can exhibit linkage isomerism?
 A. NO_2^- B. NH_3 C. H_2O D. CO
192. Which of the following ligands produces a synergic effect in coordination complexes?
 A. Oxalates B. Carbon Monoxide C. Sulphate D. Nitrates
193. Which of the following species is not a ligand?
 A. NH_4^+ B. NO C. CO D. $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$
194. Which of the following statements regarding primary and secondary valencies is correct?
 A. Primary Valencies are ionisable while Secondary Valencies are non-ionisable.
 B. Primary Valencies are non-ionisable while Secondary Valencies are ionisable.
 C. Both Primary and Secondary Valencies are non-ionisable.

D. Both Primary and Secondary Valencies are ionisable.

Application Based Questions:

195. Which type of complexes doesn't show geometrical isomerism?

A. $[MX_2L_2]$ Tetrahedral

B. $[MABXL]$ Square planar

C. $[MX_2L_2]$ Square planar

D. $[MX_4L_2]$ Octahedral

